Gaussian counterexamples to benign non-convexity in dimensional logarithmic Sobolev optimization

Principia Math In Progress

22 September 2026

Abstract

The dimensional logarithmic Sobolev majorant can have a strict local minimum which is not global, even for a centered Gaussian target in dimension two. We also give a separate Gaussian perturbation argument for the nonequivalent sign convention printed in equation (18) of the source. Thus the benign non-convexity conjecture fails under both readings.

Task 1: scope and the intended objective

Conjecture 2.6 of Li, Cui, Li, Marzouk and Zahm [1] asserts benign non-convexity on $\mathrm{Gr}(d, r)$, meaning that every local minimum is global. We take the permitted reference measure $\mu = N(0, I_d)$. For a target π, let $M = \mathbb{E}_\pi[XX^\mathsf{T}]$ and $H = \mathbb{E}_\pi[\nabla \log \pi \, \nabla \log \pi^\mathsf{T}]$. The source's equation (20) gives the majorant

$$J(U) = \frac{1}{2}\left\{\mathrm{tr}(U_\perp^\mathsf{T} M U_\perp) - (d-r) + \log\det(U_\perp^\mathsf{T} H U_\perp)\right\}, \tag{1}$$

where $[U, U_\perp]$ is orthogonal. This is also the expression derived in the source's Appendix D.2. In particular, its Proposition 4 identifies (1) with the actual approximation KL divergence for Gaussian targets. We verify that identity directly in our example below.

There is a source sign inconsistency. Appendix B correctly states

$$\det(U_\perp^\mathsf{T} H U_\perp) = \det(H)\det(U^\mathsf{T} H^{-1} U).$$

Consequently, the coefficient of $\log\det(U^\mathsf{T} H^{-1} U)$ in (1) is $+1/2$. The displayed equation (18) instead has coefficient $-1/2$. We first disprove the intended majorant landscape using (1), and then separately disprove the landscape assertion for (18) as printed. The resolution therefore does not rest on detecting the sign error.

A strict nonglobal minimum for the intended majorant

Theorem 1. *For $d = 2$, $r = 1$, $\mu = N(0, I_2)$ and*

$$\pi = N(0, C), \qquad C = \mathrm{diag}(1/2, 2),$$

both coordinate lines are strict local minima of J on $\mathrm{Gr}(2, 1)$. The line spanned by e_1 is not a global minimum.

Proof. Here $M = C$, $H = C^{-1} = \mathrm{diag}(2, 1/2)$, and $\det C = 1$. For $u(\theta) = (\cos\theta, \sin\theta)^\mathsf{T}$ put

$$z(\theta) = u(\theta)^\mathsf{T} C u(\theta) = \frac{1}{2} + \frac{3}{2}\sin^2\theta.$$

Its orthogonal complement is $v(\theta) = (-\sin\theta, \cos\theta)^\mathsf{T}$. Then $v^\mathsf{T} C v = 5/2 - z$ and $v^\mathsf{T} H v = z$, so (1) becomes

$$J(\theta) = \frac{1}{2}\left(\frac{3}{2} - z(\theta) + \log z(\theta)\right). \tag{2}$$

The angle is a local coordinate on $\mathrm{Gr}(2,1)$, with period π. At $\theta = 0$ and $\theta = \pi/2$, the first derivative vanishes, and direct differentiation gives

$$J''(0) = \frac{3}{2} > 0, \qquad J''(\pi/2) = \frac{3}{4} > 0.$$

Thus both are strict local minima in the manifold itself. Their values are

$$J(0) = \frac{1 - \log 2}{2}, \qquad J(\pi/2) = \frac{-1/2 + \log 2}{2}.$$

The difference is $3/4 - \log 2 > 0$. For an elementary exact sign proof, the graph of the strictly convex function $1/x$ lies strictly below its secant on $(1,2)$; hence

$$\log 2 = \int_1^2 \frac{dx}{x} < \int_1^2 \frac{3-x}{2}\, dx = \frac{3}{4}.$$

The other coordinate line therefore has smaller value than the strict local minimum at e_1, disproving benign non-convexity. □

These are valid probability measures under the source assumptions. Their density ratio is the smooth positive function $d\pi/d\mu = \exp(-x_1^2/2 + x_2^2/4)$, normalized under μ. All second moments, Fisher information and KL quantities used above are finite. The source assumes a smooth positive ratio, not a bounded ratio or a target covariance dominated by I_2.

To check the actual KL interpretation independently, the source's optimal profile preserves the u-marginal of π and keeps a standard Gaussian in the complementary direction. Its covariance is

$$D = I_2 + (z-1)uu^\mathsf{T}, \qquad D^{-1} = I_2 + (z^{-1} - 1)uu^\mathsf{T}, \qquad \det D = z.$$

The centered Gaussian KL formula yields

$$D_{\mathrm{KL}}(N(0,C)\|N(0,D)) = \tfrac{1}{2}\{\log z - \log\det C - 2 + \mathrm{tr}(D^{-1}C)\} = \tfrac{1}{2}(3/2 - z + \log z).$$

Thus (2) is the intended approximation loss itself, independently of the inconsistent display (18).

The literal equation (18) also has nonglobal local minima

For completeness, write the function actually printed in (18) as

$$J_{\mathrm{lit}}(u) = \frac{\mathrm{tr}\, M - 2 + \log\det H}{2} - \frac{u^\mathsf{T} M u - 1 + \log(u^\mathsf{T} H^{-1} u)}{2}. \tag{3}$$

This is again well defined on $\mathrm{Gr}(2,1)$.

Theorem 2. *Let $C = \mathrm{diag}(1/2, 2)$ and $m_\delta = (\sqrt{3}, \delta)^\mathsf{T}$. There exists $\delta_0 > 0$ such that, for every $0 < \delta < \delta_0$, the target $\pi_\delta = N(m_\delta, C)$ gives two strict local minima of (3) with different values. In particular, (3) is not benignly non-convex.*

Proof. For these targets, $H = C^{-1}$ and $M = C + m_\delta m_\delta^\mathsf{T}$, so all matrices in (3) arise from actual Gaussian probability measures. Use the local chart

$$u(t) = \frac{(1,t)^\mathsf{T}}{\sqrt{1+t^2}}, \qquad z(t) = \frac{1/2 + 2t^2}{1+t^2}.$$

Then (3), as a smooth function of (t, δ), is

$$F(t,\delta) = \frac{9/2+\delta^2}{2} - \frac{1}{2}\left\{\frac{1/2 + 2t^2 + (\sqrt{3}+\delta t)^2}{1+t^2} + \log z(t)\right\}. \tag{4}$$

At $\delta = 0$, set $w = (1+t^2)^{-1}$. Up to its additive constant, (4) equals

$$-\tfrac{1}{2}\{2 + \tfrac{3}{2}w + \log(2 - \tfrac{3}{2}w)\}.$$

Its derivative in w vanishes at $w = 2/3$, where its second derivative is $9/8 > 0$. Accordingly, the two distinct lines with $t_+ = 1/\sqrt{2}$ and $t_- = -1/\sqrt{2}$ satisfy

$$F_t(t_\pm, 0) = 0, \qquad F_{tt}(t_\pm, 0) = \frac{4}{9} > 0.$$

The implicit function theorem applied separately to $F_t = 0$ provides smooth critical points $t_\pm(\delta)$ for $|\delta|$ sufficiently small. They remain distinct strict local minima by continuity of F_{tt}. Let $v_\pm(\delta) = F(t_\pm(\delta), \delta)$ be their values. Stationarity eliminates the derivatives of $t_\pm(\delta)$, giving

$$v_\pm'(0) = F_\delta(t_\pm, 0) = -\frac{\sqrt{3}\, t_\pm}{1+t_\pm^2} = \mp\frac{\sqrt{6}}{3}.$$

At zero the two values are equal, whereas $(v_+ - v_-)'(0) = -2\sqrt{6}/3 < 0$. It follows from the definition of the derivative that $v_+(\delta) < v_-(\delta)$ for every sufficiently small positive δ. Thus the second local minimum is nonglobal, as claimed. □

Both the intended majorant and the literal printed function therefore fail the original landscape assertion. These are two readings of the same target, and receive only one problem credit.

Proposed follow-up Tasks 2–5

These are proposed tasks, not additional completed results.

2. Classify all local minima of the intended objective on $\mathrm{Gr}(d, r)$ for centered Gaussian targets with diagonal covariance.
3. Determine whether covariance domination $C \preceq I_d$ suffices for benign non-convexity for every Gaussian target, including nonzero means.
4. Give an explicit numerical threshold δ_0 in the second theorem with certified lower bounds on both local Hessians.
5. Characterize pairs of second-moment and Fisher-information matrices arising from probability measures for which (1) has no nonglobal local minima for any rank r.

References

[1] M. T. C. Li, T. Cui, F. Li, Y. Marzouk and O. Zahm, *Sharp detection of low-dimensional structure in probability measures via dimensional logarithmic Sobolev inequalities*, Information and Inference: A Journal of the IMA 14(3) (2025), iaaf021, doi:10.1093/imaiai/iaaf021. Complete author's final manuscript read from DSpace@MIT; byte-identical to arXiv:2406.13036v3.